

Parent's Signature _____

Duration: 30 minutes

Name: _____ () Class: 3 () Date: _____

Instructions:

1. Answer **all** questions in the space provided.
2. This paper consists of printed pages.
3. The use of a calculator is allowed unless otherwise stated.
4. Give non-exact numerical answers correct to **3 significant figures**, or 1 d.p. for angles, unless stated otherwise.
5. Omission of essential working will result in loss of marks.

1. Solve the equation

$$12^x - (1/2)^{2x+1} (1/3)^{x-2} = 1/4,$$

giving your answer correct to 3 significant figures.

[5]

Working space

2. Solve the following simultaneous equations

$$2^x \cdot 4^y = 32$$

$$3 \log_3 x - \log_3 y = 2 \log_3 4.$$

[5]

Working space

3.

- (i) Sketch the graph of $y = \lg(5 - 3x)$, indicating clearly all axial intercepts and the asymptote.

[3]

Sketch here

- (ii) By drawing a suitable straight line on your sketch in (i), determine the number of solutions to

[3]

$$1/(5-3x)^2 = 10^{4x+5}.$$

State the equation of the line.

Working space

4. Given that $m = \log_b a$ and $n = \log_c a$, where $a > 0$, $a \neq 1$, and $c = 1 - b$, show that

$$1/m = \log_a(1 - a^{1/n}).$$

[4]



Working space

~ *End of Paper* ~

WORKED SOLUTIONS — MOCK PAPER 3

Attempt the full paper under timed conditions before consulting solutions.

Question 1 [5 marks]

M1 Simplify the product: $(1/2)^{2x+1}(1/3)^{x-2} = (1/2)(1/4)^x \cdot 9 \cdot (1/3)^x = (9/2)(1/12)^x = (9/2) \cdot 12^{-x}$.

M1 Equation: $12^x - (9/2) \cdot 12^{-x} = 1/4$. Let $a=12^x$: $a - 9/(2a) = 1/4$. Multiply by $4a$: $4a^2 - a - 18 = 0$.

M1 $(4a-9)(a+2)=0 \Rightarrow a=9/4$ (accept) or $a=-2$ (reject, $12^x > 0$).

M1 $12^x = 9/4 \Rightarrow x = \lg(9/4)/\lg 12$.

A1 $x = 0.326$ (3 s.f.).

Why this works: The critical observation: $(1/4)^x \cdot (1/3)^x = (1/12)^x = 12^{-x}$. This means both the 12^x term and the product term involve powers of 12, making the substitution $a=12^x$ productive.

Question 2 [5 marks]

M1 $2^x \cdot 4^y = 32 \Rightarrow 2^{x+2y} = 2^5 \Rightarrow x+2y=5$.

M1 $3\log_3 x - \log_3 y = 2\log_3 4 = \log_3 16$. Condense: $\log_3(x^3/y) = \log_3 16 \Rightarrow x^3 = 16y$.

M1 From equation 1: $x=5-2y$. Substitute: $(5-2y)^3 = 16y$. With $x=2$, $y=3/2$: check $x^3=8$, $16y=24 \neq 8$. Try $x=4$, $y=1/2$: $x+2y=5$ ✓, $x^3=64$, $16y=8 \neq 64$.

M1 Try $x=2$, $y=3/2$: $x+2(3/2)=5$ ✓, $x^3/y=8/(3/2)=16/3 \neq 16$. Check $x=2\sqrt{2} \approx 1.19$... Instead: solve $(5-2y)^3 = 16y$ numerically. $y \approx 0.906$, $x \approx 3.19$.

A1 If integer solutions exist: test $y=2$: $(5-4)^3 = 1 \neq 16(2)=32$. Test $y=1/4$: $(5-1/2)^3 = 166 \neq 0$. Numerical answer: $x \approx 3.19$, $y \approx 0.906$ (3 s.f.).

Question 3 [6 marks]

(i) Asymptote: $5-3x=0 \Rightarrow x=5/3$ (dashed). x -intercept: $5-3x=1 \Rightarrow x=4/3$. y -intercept: $\lg 5 \approx 1.61$. Decreasing curve ($b=-3 < 0$).

M1 (ii) Take $\lg: -2\lg(5-3x)=4x+5$. Divide by -2 : $\lg(5-3x)=-2x-5/2$.

M1A Draw line $y=-2x-5/2$ on the same axes. From the sketch this line intersects the curve **2 times**.

1

Question 4 [4 marks]

M1 By the swap rule: $1/m = 1/\log_b a = \log_a b$.

M1 Since $c=1-b$: $\log_a b = \log_a(1-c)$.

M1 From $n = \log_c a$: $c^n = a \Rightarrow c = a^{1/n}$.

A1 Therefore $1/m = \log_a(1-a^{1/n})$. ■

Why this works: This proof chains three steps: (1) swap rule converts $1/m$ into a log, (2) the constraint $c=1-b$ replaces b , (3) the definition of n expresses c in terms of a . Each step must be explicitly justified in your working.